

MOHAMMED MASUD CHOWDHURY MAHIR

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SUMMARY

Motivated and detail-oriented Cybersecurity enthusiast with a strong foundation in vulnerability assessment, threat analysis, and secure coding practices. Skilled in using security tools such as Burp Suite, Nmap, Wireshark, Metasploit, and Ghidra for practical testing and analysis. Proficient in Python, C/C++, Bash, SQL databases, and modern development workflows with Git, Docker, FastAPI, PyTorch, and TensorFlow. Able to analyze complex technical problems, implement secure solutions, and adapt quickly to new technologies while contributing effectively to security-focused projects.

SKILLS

Languages: Python, C/C++, Bash, PHP

Database: SQL(Sqlite, MySQL), Excel

Frameworks: PyTorch, TensorFlow, FastAPI

Methodologies: OWASP Top 10, Vulnerability Assessment, Penetration Testing

Reporting & Compliance: : Technical Writing, ISO 27001 Awareness, Risk Analysis

Developer Tools: Git, Github, Docker

Security Tools: Burp Suite, Nmap, Wireshark, Metasploit, Ghidra

Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Transformer, Huggingface

EXPERIENCE

Senior Executive & Trainer

Apr. 2024 – Present

UITS COMPUTER CLUB - A university club under supervised the department of CSE and other tech leaders.

UITS, Dhaka

- Lead cybersecurity training sessions and mentor students in Capture the Flag (CTF) challenges.
- Conduct workshops on networking, threat detection and web application security.
- Organize and oversee cybersecurity events, competitions, and awareness programs.
- Created detailed technical documentation and skill-building roadmaps to guide members through complex security concepts.
- UITS Skillathon CTF 2025 - Organizer
- UITS Hacked 101 CTF - Organizer
- UITS Winter CTF Junior - Organizer

PROJECTS

Social Engineering Defense AI (SED-AI) | Python, LangChain, FAISS, Sentence Transformers, Ollama, Gemma

Feb. 2026

- Developed a fully offline RAG-based AI assistant for phishing and social engineering defense training.
- Processed and indexed multiple cybersecurity PDFs into a FAISS vector database for fast semantic retrieval.
- Integrated Sentence Transformers embeddings to enable accurate context-aware document search.
- Deployed Gemma LLM via Ollama for local response generation without external APIs or cloud dependency.
- Built an interactive Gradio chatbot interface with source citations (PDF + page references) for trustworthy answers.
- Implemented ethical safety guardrails to refuse harmful or offensive social engineering requests.

BDTour-LLM | Python, Hugging Face, Unsloth, LoRA, Qwen-2.5, Mistral, LLaMA, PEFT

Jan. 2026

- Engineered BDTour-LLM, a specialized conversational AI fine-tuned for high-accuracy travel guidance and deep cultural insights specific to the Bangladesh tourism sector.
- Optimized fine-tuning workflows using Unsloth and LoRA, achieving a 2x speedup in training and reducing VRAM overhead by 60% compared to standard HF scripts.
- Curated and pre-processed a localized dataset of tourism benchmarks, implementing Supervised Fine-Tuning (SFT) to enhance model performance on niche regional queries.
- Developed scalable NLP pipelines using Hugging Face Transformers to deliver context-aware responses with reduced hallucination rates in low-resource language contexts.

AI Summarizer | Python, Hugging Face Transformers, PyTorch, CNN/DailyMail, t5-small, ROUGE, Seq2Seq

Jan. 2026

- Developed a T5-based abstractive text summarization system fine-tuned on the CNN/DailyMail dataset using Hugging Face Transformers.

- Optimized the training pipeline with Seq2SeqTrainer, gradient accumulation, FP16 training, and custom checkpoint management for efficient GPU usage.
- Achieved ROUGE-1, ROUGE-2, and ROUGE-L scores of 0.41, 0.187, and 0.289 respectively after 2.5k training steps.
- Implemented a robust evaluation workflow including ROUGE-based metric computation, decoding optimization, and visualization of loss/metric curves.
- Built a custom summary generation function enabling high-quality inference with beam search, repetition penalty, and length control.

AmarBangla-whisper | *Python, Hugging Face Transformers, PyTorch, Librosa, Pandas,whisper-small* Nov. 2025

- Developed AmarBangla-whisper, an AI-powered ASR system fine-tuned for Bengali regional dialects in the Shobdotori Competition.
- Achieved a Levenshtein Similarity Score of 0.88573, demonstrating strong robustness across 20 dialects of Bangladesh.
- Performed extensive data cleaning and quality checks on large audio datasets using Python (Pandas) to ensure high-accuracy outputs.

VOROSHA-Web-App | *Python, Django, SQLite, REST API, HTML, CSS, JavaScript* May 2024

- Developed a Django-based web application to streamline blood donation and donor–recipient management.
- Implemented secure authentication, donor availability tracking, and automated blood request workflows.
- Designed and managed a relational SQLite database schema to track donor availability and ensure accurate record-keeping.
- Built a responsive and user-friendly interface using HTML, CSS, and JavaScript.

CERTIFICATIONS

Certified Web App Penetration Testing Apprentice (kWAPTA) - Knight Squad Academy

Issued Jan 2026

Certified Network Security Practitioner(CNSP) - The SecOps Group

Issued Aug 2025

EDUCATION

University of Information Technology and Sciences Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering (BSC) 2022 – 2026
 Current CGPA: 3.45 / 4.00 (Tentative)

Cambrian College Dhaka, Bangladesh
Higher Secondary Certificate (HSC), Science 2020
 GPA: 5.00 / 5.00

Adamjee Cantonment Public School Dhaka, Bangladesh
Secondary School Certificate (SSC), Science 2018
 GPA: 4.89 / 5.00

EXTRACURRICULAR ACTIVITIES

Hackify FEST CTF 2025 2025
Champion

UITS Winter CTF 2024 2024
1st Runner-up

UITS Next Gen Leadership CTF 2024 2024
1st Runner-up

UITS Idea Contest 2024 2024
1st Runner-up